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Unit – 3 ⇝ Linear Transformation (Lecture Note)

Introduction

→ **Linear Transformation** is a mapping between vector spaces that preserves vector addition and scalar multiplication.

→ Linear transformations can be represented using **matrices**, which simplify computations and analysis.

→ The **kernel**, **range**, and **Rank–Nullity Theorem** help us understand the properties of a linear transformation.

→ Different matrix representations of the same transformation can be obtained by changing the **basis**, leading to the concept of **similar matrices.**

→ Linear transformations have important applications in Artificial Intelligence, Machine Learning, computer graphics, image processing, and data analysis**.**

Method 1 ⇝ Linear Transformations

Linear Transformation

→ Let V and W be two vector spaces. A **linear transformation** T is a mapping (function) from V to W ($T: V \to W$) such that

(1) $T(u+v) = T(u) + T(v)$

(2) $T(ku) = k\ T(u)$

for all vectors $u, v \in V$ and scalars $k \in \mathbb{R}$.

→ If $T: V \to W$ be a linear transformation and $u \in V$, then $T(u)$ is called the **image** of u under T.

- **Example:** Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation defined as,

  $T(x, y) = (x + y, x - y)$ then $T(1, 0) = (1 + 0, 1 - 0) = (\mathbf{1}, \mathbf{1})$.

  So, $(1, 1)$ is **image** of element $(1, 0)$ under the transformation T.

→ If $V = W$, the linear transformation $T: V \to V$ is called **linear operator**.

→ **Procedure to Check Mapping T is a Linear Transformation or Not**

Let $T: V \to W$ be any mapping.

(1) Select two elements u and v from the V and one scalar $k \in \mathbb{R}$.

(2) Find $T(u+v), T(u) + T(v), T(ku)$, and $kT(u)$.

(3) Verify $T(u+v) = T(u) + T(v)$ and $T(ku) = kT(u)$.

(4) If both conditions are satisfied, then the given mapping T is a linear transformation otherwise not.

Some Standard Linear Transformations

(1) **Zero Transformation**

- Let V and W be two vector spaces. The function $T: V \to W$ defined by $\mathbf{T(v)} = \mathbf{0}$; $\forall\ v \in V$ is a zero linear transformation from V to W.

  Let $u, v \in V$ and $k \in \mathbb{R}$

  $\Rightarrow T(u) = 0,\ \ T(v) = 0 \Rightarrow T(u) + T(v) = 0$

  $T(u+v) = 0 = T(u) + T(v)$ and $\quad (\because u + v \in V)$

  $T(ku) = 0 = k\,T(u) \quad (\because ku \in V)$

  $\Rightarrow T(u+v) = T(u) + T(v)$ and $T(ku) = k\,T(u)$.

- Hence, T is a linear transformation and it is called zero transformation.

(2) **Identity Operator**

- Let V be a vector space. The function $I: V \to V$ defined by $\mathbf{I(v) = v};\ \forall\ v \in V$ is a linear operator from V to V.

  Let $u, v \in V$ and $k \in \mathbb{R}$

  $\Rightarrow I(u) = u,\ \ I(v) = v \Rightarrow I(u) + I(v) = u + v$

  $I(u + v) = u + v = I(u) + I(v)$ and $(\because u + v \in V)$

  $I(ku) = ku = k\,I(u)$ $(\because ku \in V)$

  $\Rightarrow T(u + v) = T(u) + T(v)$ and $T(ku) = k\,T(u)$

- Hence, I is linear transformation and it is called identity operator on V.

→ **Properties of Linear Transformations**

If $T: V \to W$ is a linear transformation, then

(1) $T(0) = 0$

(2) $T(-v) = -T(v)$ for all $v \in V$.

(3) $T(v - w) = T(v) - T(w)$ for all $v, w \in V$.

(4) $T(k_1v_1 + k_2v_2 + \cdots + k_nv_n) = k_1T(v_1) + k_2T(v_2) + \cdots + k_nT(v_n)$.

where, $v_1, v_2, \ldots v_n$ are vectors in V and $k_1, k_2, \ldots . k_n$ are scalars in $\mathbb{R}$.

Examples of Method-1: Linear Transformations

C	1	Show that the following functions are linear transformations. (i) $T: \mathbb{R}^2 \to \mathbb{R}^2$, where $T(x, y) = (x + 3y,\ 3x - y)$. (ii) $T: \mathbb{R}^2 \to \mathbb{R}^2$, where $T(x, y) = (2x - y,\ x - y)$.
C	2	Check whether $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by, $T(x_1, x_2, x_3) = (x_1 + 1,\ x_2 + 2,\ x_3 + 4)$ is a linear transformation or not? **Answer: No**
C	3	Check whether $T: \mathbb{M}_{2\times2} \to \mathbb{R}$ defined by $T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a^2 + b^2$ is a linear transformation or not? **Answer: No**

C	4	Define a function $T : \mathbb{P}_2 \to \mathbb{P}_2$ by $T(p(x)) = p(x+1)$. Show that T is a linear transformation. Also find the image of the polynomial $q(x) = 3x^2 - x + 2$. **Answer : $q(x) = 3x^2 + 5x + 4$**

Method 2 ⇝ Kernel, Range and Rank-Nullity Theorem

Standard Matrix of the Linear Transformation

→ If $T : V \to W$ is the linear transformation defined in the form of $T(x) = Ax$ for every $x \in V$, then the matrix A is called the **standard matrix** of T.

→ It is denoted by **[T].**

→ If $T : V \to W$ be a linear transformation with $\dim(V) = n$ and $\dim(W) = m$, then order of [T] is $\mathbf{m \times n}$.

→ The standard basis of V is $\{e_1, e_2, \dots, e_n\}$ and the standard basis of W is $\{e'_1, e'_2, \dots, e'_m\}$.

→ **Procedure to Find the Standard Matrix [T]**

(1) Write the standard basis $\{e_1, e_2, \dots, e_n\}$ of the domain V.

(2) Find $T(e_1), T(e_2), \dots, T(e_n)$.

(3) Write each $T(e_1), T(e_2), \dots, T(e_n)$ into the **linear combination** of the vectors $e'_1, e'_2, \dots, e'_m$.

i.e.,
$$T(e_1) = k_1^1 e'_1 + k_2^1 e'_2 + \cdots + k_m^1 e'_m$$
$$T(e_2) = k_1^2 e'_1 + k_2^2 e'_2 + \cdots + k_m^2 e'_m$$
$$\vdots$$
$$T(e_n) = k_1^n e'_1 + k_2^n e'_2 + \cdots + k_m^n e'_m.$$

(4) Write $k_1^i, k_2^i, \dots, k_m^i$ into the ith column of the matrix for each $i \in \{1, 2, \dots, n\}$.

(5) The resulting matrix $[T] = \begin{bmatrix} k_1^1 & \cdots & k_1^n \\ \vdots & \ddots & \vdots \\ k_m^1 & \cdots & k_m^n \end{bmatrix}$ is the standard matrix of T.

Kernel and Range

→ Let $T : V \to W$ be a linear transformation. The **kernel** or **null space** of T is the set of all vectors in V that maps to the zero vector of W.

→ It is denoted by ker(T) or N(T).

i. e. $\ker(T) = \{v \in V : T(v) = 0\}$.

→ Let $T : V \to W$ be a linear transformation. The **Range Space** of T is the set of all vectors in W that are images of at least one vector in V under T.

→ It is denoted by R(T).

i. e. $R(T) = \{T(v) \in W : v \in V\}$.

→ **Remarks**:

(1) ker(T) and R(T) are always subspaces of V and W respectively.

(2) If $T : V \to W$ be a zero transformation, then $\mathbf{ker(T) = V}$ and $\mathbf{R(T) = \{0\}}$.

(3) If $I : V \to V$ be an identity operator, then $\mathbf{ker(I) = \{0\}}$ and $\mathbf{R(I) = V}$.

Rank and Nullity of Linear Transformation

→ Let $T : V \to W$ be a linear transformation. The **rank** of T is the dimension of the range space of T.

→ It is denoted by rank(T).

i. e. $\text{rank(T)} = \dim\big(\text{R(T)}\big)$.

→ Let $T : V \to W$ be a linear transformation. The **nullity** of T is the dimension of the kernel of T.

→ It is denoted by nullity(T).

i. e. $\text{nullity(T)} = \dim\big(\ker\text{(T)}\big)$.

→ **Procedure to Find the Rank of T**

Let $T : V \to W$ be a linear transformation.

(1) Find the standard matrix [T] of given linear transformation.

(2) Find the column space of standard matrix [T].

(3) Find the basis of column space which is also the basis of R(T).

(4) Number of elements in basis of $\text{R(T)} = \dim\big(\text{R(T)}\big) = \mathbf{rank(T)}$.

→ **Procedure to Find the Nullity of T**

Let $T : V \to W$ be a linear transformation.

(1) Construct homogeneous system $\mathbf{T(u) = 0}$ from the given linear transformation.

(2) Find the solution of homogeneous system and write down the solution space.

(3) Find the basis for $\mathbf{ker(T)}$ from the solution space.

(4) Number of elements in basis of $\ker\text{(T)} = \dim\big(\ker\text{(T)}\big) = \mathbf{nullity(T)}$.

Rank-Nullity Theorem

→ If $T : V \to W$ is a linear transformation from a finite dimensional vector space V to W, then $\mathbf{rank(T) + nullity(T) = dim(V)}$.

→ **Remarks:**

(1) rank(T) = rank of [T] = ρ([T])

(2) nullity(T) = nullity of [T] = η([T])

Examples of Method-2: Kernel, Range and Rank-Nullity Theorem

C	1	Find the standard matrix of $T: \mathbb{R}^3 \to \mathbb{R}^4$ defined by $T(x, y, z) = (x + y + z, x + 2y - 3z, 2x + 3y - 2z, 3x + 4y - z)$. **Answer:** $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & -2 \\ 3 & 4 & -1 \end{bmatrix}$
C	2	Let $T : \mathbb{P}_1 \to \mathbb{P}_2$ be the linear transformation defined by $T(p(x)) = x(p(x))$. Find the matrix [T]. **Answer:** $\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$
C	3	If $T : \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ is a linear transformation defined by $T(A) = MA$, where $M = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$. Find the standard matrix of T relative to the standard basis. **Answer:** $\begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ -1 & 0 & 3 & 0 \\ 0 & -1 & 0 & 3 \end{bmatrix}$
C	4	Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator defined by $T(x, y) = (2x - y, -8x + 4y)$ then find the bases of ker(T) and R(T). **Answer:** **Basis of ker(T)** $= \left\{\left(\frac{1}{2}, 1\right)\right\}$, **Basis of R(T)** $= \{(2, -8)\}$.

C	5	Let $T: \mathbb{P}_2 \to \mathbb{P}_3$ be the linear transformation defined by $T(p(x)) = x \cdot p(x)$ then find the bases of ker(T) and R(T). **Answer: There is no basis for the ker(T).** **Basis of R(T) $= \{x, x^2, x^3\}$.**
C	6	Let $T: \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ be the linear transformation defined by $T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+b & b+c \\ a+d & b+d \end{bmatrix}$ then find the bases of ker(T) and R(T). Also find rank(T) and nullity(T). **Answer: There is no basis for the ker(T).** **Basis of R(T) $= \left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\right\}$** **nullity(T) $= 0$, rank(T) $= 4$.**
C	7	Verify rank-nullity theorem for the linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (x + y + z, x + y)$.

Method 3 ⇝ Matrix Representation of Linear Transformations

Associated Matrix of Linear Transformation with Respect to Basis

→ Let $T : V \to W$ be a linear transformation.

→ If S_1 is a basis of vector space V and S_2 is a basis of vector space W, then the matrix associated to T with respect to $\mathbf{S_1}$ & $\mathbf{S_2}$ is denoted by

$$[\mathbf{T}]_{\mathbf{S_2},\mathbf{S_1}} = [\mathbf{T}]_{\textbf{codomain basis, domain basis}}$$

→ **Procedure to Find the Matrix Associated to Linear Transformation T**

Let $S_1 = \{v_1, v_2, \dots, v_n\}$ is basis of V and $S_2 = \{w_1, w_2, \dots, w_m\}$ is basis of W.

(1) Find $T(v_1), T(v_2), \dots, T(v_n)$.

(2) Each element of W can be written in the linear combination of the elements of set S_2. So, write obtained elements as follows.

$$T(v_1) = k_1^1 w_1 + k_2^1 w_2 + \cdots + k_m^1 w_m$$
$$T(v_2) = k_1^2 w_1 + k_2^2 w_2 + \cdots + k_m^2 w_m$$
$$\vdots$$
$$T(v_n) = k_1^n w_1 + k_2^n w_2 + \cdots + k_m^n w_m$$

(3) Write $k_1^i, k_2^i, \dots, k_m^i$ into the ith column of the matrix for each $i \in \{1, 2, \dots, n\}$.

(4) The resulting matrix $[T]_{S_2,S_1} = \begin{bmatrix} k_1^1 & \cdots & k_1^n \\ \vdots & \ddots & \vdots \\ k_m^1 & \cdots & k_m^n \end{bmatrix}$ is the **matrix associated to T** with respect to S_1 and S_2.

→ **Remarks:**

(1) If $T : V \to W$ is a linear transformation with $\dim(V) = n$ and $\dim(W) = m$, then the matrix associated to T is of order $\mathbf{m \times n}$.

(2) If S_1 and S_2 are the standard bases of V and W respectively, then $[\mathbf{T}]_{\mathbf{S_2},\mathbf{S_1}} = [\mathbf{T}]$.

→ **Matrices of Identity Operators**

→ Let $T : V \to V$ be a linear operator. If $S_1 = \{v_1, v_2, \dots, v_n\}$ and $S_2 = \{w_1, w_2, \dots, w_n\}$ are two different non-standard bases of V, then

- $[T]_{S_1}$ denotes the matrix of T when the domain and codomain use the **same basis S_1.**
- $[T]_{S_2}$ denotes the matrix of T when the domain and codomain use the **same basis S_2.**

- $[T]_{S_2,S_1}$ denotes the matrix of T with respect to basis S_1 of the **domain** and basis S_2 of the **codomain.**
- $[T]_{S_1,S_2}$ denotes the matrix of T with respect to basis S_2 of the **domain** and basis S_1 of the **codomain.**

Examples of Method-3: Matrix representation of a Linear Transformation

C	1	Let $T: \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation defined by $T(x,y) = (y, -5x+13y, -7x+16y)$. Find the matrix of the transformation T with respect to the bases $S_1 = \{v_1, v_2\}$ for $\mathbb{R}^2$ and $S_2 = \{w_1, w_2, w_3\}$ for $\mathbb{R}^3$, where $v_1 = (3,1), v_2 = (5,2)$, $w_1 = (1,0,-1), w_2 = (-1,2,2)$ and $w_3 = (0,1,2)$. **Answer:** $[T]_{S_2,S_1} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -2 & -1 \end{bmatrix}$
C	2	Let $T: \mathbb{P}^1 \to \mathbb{P}^2$ be the defined by $T(p(x)) = x \cdot (p(x))$. Find the matrix of the transformation T with respect to the bases $S_1 = \{v_1, v_2\}$ and $S_2 = \{w_1, w_2, w_3\}$, where $v_1 = 1, v_2 = x, w_1 = x+1, w_2 = x-1$ and $w_3 = x^2$. **Answer:** $[T]_{S_2,S_1} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$
C	3	Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x,y,z) = (x+2y+z, 2x-y, 2y+z)$. Find the matrix of transformation T with respect to (i) S_1, (ii) S_1 & S_2, (iii) S_2 & S_1, (iv) S_2 where, $S_1 = \{(1,0,0), (0,1,0), (0,0,1)\}$ and $S_2 = \{(1,0,1), (0,1,1), (0,0,1)\}$. **Answer:** (i) $[T]_{S_1} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$, (ii) $[T]_{S_2,S_1} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$ (iii) $[T]_{S_1,S_2} = \begin{bmatrix} 2 & 3 & 1 \\ 2 & -1 & 0 \\ 1 & 3 & 1 \end{bmatrix}$, (iv) $[T]_{S_2} = \begin{bmatrix} 2 & 3 & 1 \\ 2 & -1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$

C	4	Let $T : \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ be the linear transformation defined by $T(A) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} A$, then find the matrix of the transformation $[T]_{Y,X}$ with respect to the standard basis X and a non-standard basis $Y = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\right\}$. **Answer**: $[T]_{Y,X} = \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{2} & \mathbf{-2} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{2} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Method 4 ⇝ Change of Basis

Effect of Change of Bases on Linear Operators

→ Let $T : V \to V$ is a linear operator.

→ The matrix of T depends on the basis chosen for V.

→ If we change the basis, the matrix representation of T also changes, but the operator T itself remains the same.

→ **Procedure to Find $[T]_{S_2}$ from $[T]_{S_1}$**

Let $S_1 = \{v_1, v_2, \dots, v_n\}$ and $S_2 = \{w_1, w_2, \dots, w_n\}$ be two bases of V.

(1) Find $[T]_{S_1}$ if not given.

(2) Take w_1 from S_2 and express as a linear combination of the vectors of S_1.

i.e., $w_1 = a_1v_1 + a_2v_2 + \cdots + a_nv_n$

(3) Write the coefficients as a column vector.

i.e., $[w_1]_{S_1} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

(4) Repeat the same process for $w_2, w_3, \dots, w_n$ and find $[w_2]_{S_1}, [w_3]_{S_1} \dots [w_n]_{S_1}$.

(5) Put these vectors into columns in order

$$P = [\,[w_1]_{S_1} \quad [w_2]_{S_1} \quad \cdots \quad [w_n]_{S_1}]$$

(6) The resultant matrix P of order $n \times n$ is called the **transition matrix** from S_2 **to** S_1**.**

(7) Find P^{-1} which is the **transition matrix** from S_1 **to** S_2**.**

(8) Find $[T]_{S_2} = P^{-1}\,[T]_{S_1}\,P$**.**

Examples of Method-4: Change of Basis

C	1	Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x,y) = (2x+y, x-3y)$. $S_1 = \{(1,2),(1,3)\}$ and $S_2 = \{(-1,1),(0,1)\}$. (i) Find the matrix of T with respect to the basis S_1. (ii) Find the matrix of T with respect to the basis S_2 by using transition matrix P. **Answer:** $[T]_{S_1} = \begin{bmatrix} 17 & 23 \\ -13 & -18 \end{bmatrix}$, $[T]_{S_2} = \begin{bmatrix} 1 & -1 \\ -5 & -2 \end{bmatrix}$
C	2	Let $S_1 = \{v_1, v_2, v_3\}$ be a basis of a vector space V and $T: V \to V$ be a linear operator such that $[T]_{S_1} = \begin{bmatrix} -3 & 4 & 7 \\ 1 & 0 & -2 \\ 0 & 1 & 0 \end{bmatrix}$. Find $[T]_{S_2}$ where $S_2 = \{w_1, w_2, w_3\}$ is the basis for V defined by $w_1 = v_1$, $w_2 = v_1 + v_2$, $w_3 = v_1 + v_2 + v_3$. **Answer:** $[T]_{S_2} = \begin{bmatrix} -4 & 0 & 9 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$
C	3	Let $S_1 = \{v_1, v_2, v_3\}$ and $S_2 = \{w_1, w_2, w_3\}$ be the bases for a vector space V and let $P = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 1 & 4 \\ 0 & 1 & 2 \end{bmatrix}$ be the transition matrix from S_2 to S_1. (i) Express w_1, w_2, w_3 as linear combinations of v_1, v_2, v_3. (ii) Express v_1, v_2, v_3 as linear combinations of w_1, w_2, w_3. **Answer:** $w_1 = 2v_1 + v_2$, $w_2 = -v_1 + v_2 + v_3$, $w_3 = 3v_1 + 4v_2 + 2v_3$ $v_1 = -2w_1 - 2w_2 + w_3$, $v_2 = 5w_1 + 4w_2 - 2w_3$, $v_3 = -7w_1 - 5w_2 + 3w_3$.

Method 5 ⇝ Similarity of Matrices

Similarity of Matrices

→ If A and B are two square matrices of order n, then B is said to be similar to A, if there exists a **non-singular** matrix P such that $B = P^{-1}AP$.

→ If A and B are two square matrices of order n, then B is said to be orthogonally similar to A, if there exists an **orthogonal** matrix P' such that $B = P'^{-1}AP'$.

→ Since P' is orthogonal, $P'^{-1} = P'^{T}$.

So, $B = P'^{-1}AP' = P'^{T}AP'$

→ **Properties of Similarity of Matrices**

(1) Similar matrices have the same **trace, determinant, rank** and **nullity.**

(2) Every **real** symmetric matrix is **orthogonally similar** to a diagonal matrix.

→ **Remarks:**

Let $T : V \to V$ be a linear operator and S_1 and S_2 are two different non-standard bases for the vector space V.

(1) The matrices $[T]$, $[T]_{S_1}$ and $[T]_{S_2}$ are **similar**.

- $\text{tr}([T]_{S_1}) = \text{tr}([T]) = \text{tr}([T]_{S_2})$
- $\det([T]_{S_1}) = \det([T]) = \det([T]_{S_2})$
- $\rho([T]_{S_1}) = \rho([T]) = \rho([T]_{S_2})$
- $\eta([T]_{S_1}) = \eta([T]) = \eta([T]_{S_2})$

(2) Trace, determinant, rank and nullity of $[T]_S$ does not depend on the choice of basis S. It depends only on the linear operator T.

Examples of Method-5: Similarity of Matrices

C	1	Prove that if A is similar to B and B is similar to C, then A is similar to C.
C	2	Show that the matrices $A = \begin{bmatrix} 3 & 1 & 3 \\ 6 & 2 & 6 \\ -3 & -1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & -3 \\ 6 & 5 & -2 \end{bmatrix}$ are not similar. **Hint: $\rho(A) \neq \rho(B)$**
C	3	Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x, y, z) = (x + 2y - z, -y, x + 7z)$. Suppose S_1 is the standard basis for $\mathbb{R}^3$ and $S_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$. Verify that $\det([T]_{S_1}) = \det([T]_{S_2})$.

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